

1. (10%) If given three nodes A, B, C:
 - (1) How many different binary trees can be constructed?
 - (2) How many different ordered trees can be constructed?
 - (3) How many different unordered trees (also known as oriented trees) can be constructed?
 - (4) How many different free trees (i.e., connected acyclic graphs) can be constructed?
 - (5) If one of the pre-order traversal, in-order traversal, or post-order traversal of the three nodes is ABC, how many different binary trees can be constructed?
2. (a) (5%) Please represent the following tree as a binary tree.
 (b) (5%) Write the post-order form for the following binary tree of arithmetic expression.

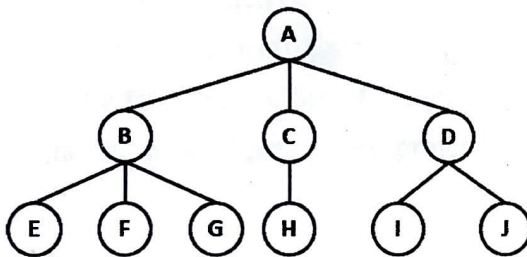


Fig. (a)

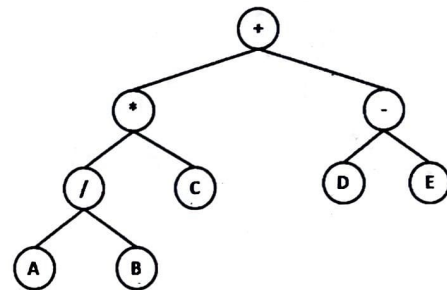


Fig. (b)

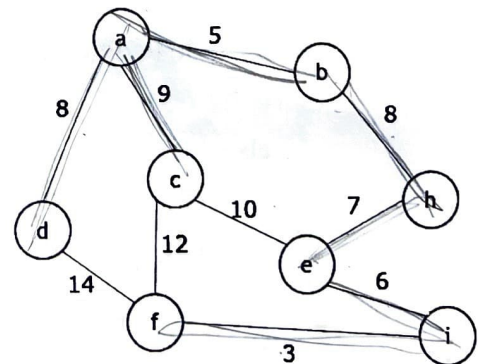
- (c) (5%) Based on the results of in-order traversal and post-order traversal, draw the binary tree.

In-order : A B C D E F G H I J K

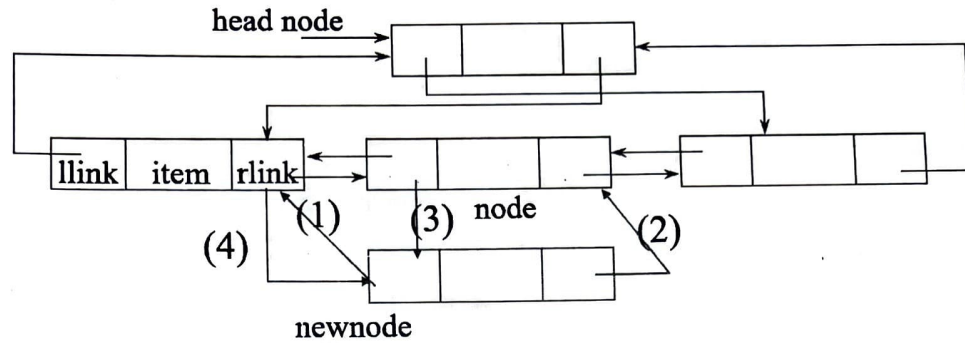
Post-order : A C E D B I H K G F

3. For the given graph as shown on the right.

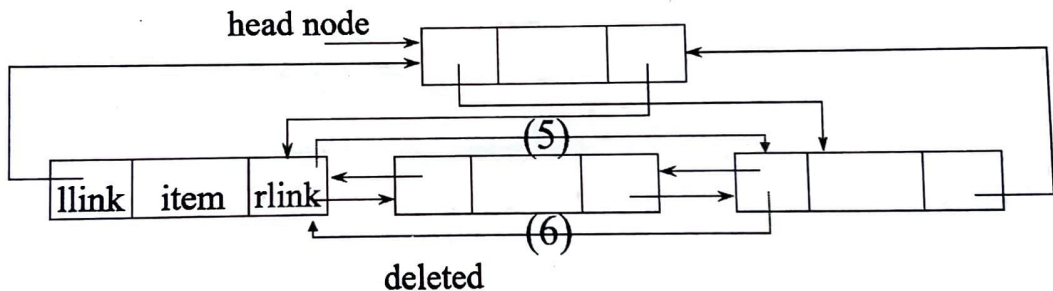
- (a) (15%) Please use the Kruskal's, Prim's, and Sollin's algorithms to find the minimum cost spanning tree. Please provide detailed explanations for step by step.
- (b) (10%) Write pseudo-code to describe how to obtain the result of the Kruskal's algorithm for (a).



4. (24%) Complete the functions for inserting and deleting nodes in a doubly circular linked list, as shown in the diagram below.



```
void dinsert(node_pointer node, node_pointer newnode)
{
    (1);
    (2);
    (3);
    (4);
}
```



```
void ddelete(node_pointer node, node_pointer deleted)
{
    if (node==deleted) printf("Deletion of head node
                             not permitted.\n");

    else {
        (5)
        (6)
        free(deleted);
    }
}
```

5. (10%) Write a function `int range_count (struct binarytree*rootPtr,int a,int b)` which accepts a pointer `rootPtr` to the root of a binary search tree `T` and returns the number of nodes in `T` having key values between `a` and `b` ($a \leq \text{key} \leq b$). The node structure of the binary search tree is defined as

```
struct binarytree {
    int key;
    binarytree *left, *right;
};
```

You may call other user-defined functions in the function `range_count`.

6. (6%) Please design a program to add a new node 't' to the right of node 's' in a threaded binary tree.
7. (15%) For the AOE (activity on edge) network shown below, do the following
- Compute the early starting time for each activity.
 - Compute the late starting time for each activity.
 - Determine the critical path of the project.

